Projective Reed–Solomon Syndromes Beyond Redundancy Four: Complete First Levels and Coherent Polar Flags

Tavis Rudd

Abstract

We classify, up to projective-semilinear equivalence, the one-column maximum-distance-separable extensions of projective Reed–Solomon codes at redundancy five over every prime power $q \geq 7$, and give exact projective counts. Equivalently, we classify the deep-hole syndrome directions in the first case left open by the classification through redundancy four. A syndrome determines a Hankel linear system of binary forms and is deep exactly when that system contains no completely split squarefree member. The generic redundancy-five case is controlled by an integral off-diagonal fiber-square curve of arithmetic genus one.

For higher redundancy we introduce coherent polar flags: iterated contractions in which every selected factor remains a marked forbidden root. Separating contained components from transverse point counts gives complete all-field syndrome classifications at redundancies six and seven; the redundancy-seven split-free list is a deep-hole classification whenever the covering radius is six, in particular for $q \geq 11$.

Index Terms

Algebraic coding theory, deep holes, finite fields, maximum-distance-separable extensions, normal rational curves, projective Reed–Solomon codes.

I. Introduction

This paper has two principal contributions. First, it gives the complete all-field classification at redundancy five, the first case left open by the known classification through redundancy four. Second, it introduces coherent polar flags: a marked contraction method that propagates splitting information to higher redundancies without forgetting the factor needed for a squarefree lift. This gives complete all-field syndrome classifications at redundancies six and seven. We use deep-hole syndromes because their Hankel systems make both the base classification and the propagation mechanism visible.

Let $\mathrm{PRS}(k)$ denote the projective Reed–Solomon code of length $q+1$ obtained by evaluating homogeneous binary forms of degree $k-1$ on $\mathbb{P}^1(\mathbb{F}_q)$. If the redundancy is

$$r = q + 1 - k,$$

a parity-check matrix has as columns the $\mathbb{F}_q$-points of the degree-$(r-1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \ldots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \ldots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

Call a projective syndrome direction split-free if it is not contained in the span of $r-2$ distinct columns. When $\rho(\mathrm{PRS}(q+1-r)) = r-1$, the split-free directions are precisely the projective syndrome directions of deep-hole cosets. The logical separation used throughout the paper is

$$\boxed{\text{split-free classification} \;+\; \rho(\mathrm{PRS}(q+1-r)) = r-1 \quad \Longrightarrow \quad \text{deep-hole classification.}}$$

The first term is proved geometrically; the covering-radius equality is a separate coding-theoretic input and is never inferred from a finite split-free census.

The correspondence between deep holes, one-symbol MDS extensions, and uncovered points of a normal rational curve was developed in [1], [2]. Zhang, Wan, and Kaipa classified projective Reed–Solomon deep-hole directions through redundancy four and identified redundancy five as the next case [2]. Our first theorem gives the corresponding all-field redundancy-five classification.

Theorem I.1 (Complete redundancy-five classification). Let $q \geq 7$ be a prime power. The covering radius of $\mathrm{PRS}(q-4)$ is 4. Its projective deep-hole syndromes are exactly the following disjoint families:

(i) the tangent family T, of size $q(q+1)$;
(ii) the conjugate-secant family S, of size $q(q^2-1)/2$;
(iii) if $\operatorname{char} \mathbb{F}_q \neq 3$, the rational osculating-pair family O^+ when $q \equiv 2 \pmod 3$, of size $q(q+1)/2$, or the conjugate osculating-pair family O^- when $q \equiv 1 \pmod 3$, of size $q(q-1)/2$;
(iv) in characteristic three, one $\mathrm{PGL}_2(q)$-fixed nucleus point and one wild Artin–Schreier orbit W of size $(q^2-1)/2$;

(v) the sporadic $\mathrm{PGL}_2(q)$-orbits in Table IV, occurring exactly for $q \in \{7, 8, 9, 11, 13, 17, 19\}$.

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2}, \qquad \frac{q(q+1)(q+2)}{2}, \qquad \frac{q^3+3q^2+q+1}{2},$$

according as $q \equiv 1 \pmod 3$ with characteristic not three, $q \equiv 2 \pmod 3$, or $\operatorname{char} \mathbb{F}_q = 3$.

The main mechanism is a characteristic-free Hankel dictionary. A syndrome at redundancy r annihilates a linear system of binary forms of degree $r-2$. It is split-free if and only if this system contains no completely split squarefree form; it is a deep-hole direction when, in addition, $\rho(\mathrm{PRS}(q+1-r)) = r-1$. At redundancy five the system is a pencil of cubics. Its induced degree-three map of projective lines is either cyclic or has geometric monodromy S_3. Cyclic descent gives the arithmetic families in Theorem I.1; in the S_3 case the off-diagonal fiber square is an integral curve of arithmetic genus one, and the singular-curve Weil bound [3] forces a split fiber for $q \geq 23$. The certified finite computations in Section VII handle the remaining fields.

Starting at redundancy six, the Hankel system is no longer a pencil. The useful replacement is the coherently parameterized line of contractions obtained by choosing one prospective root. Iteration produces a polar flag. Geometrically, naïve contraction keeps the lower divisor but forgets which point of $\mathbb{P}^1$ was removed. If the lower divisor passes through that point, putting the point back creates a double point rather than a squarefree divisor. The polar flag retains the removed point as a forbidden marker, exactly the datum needed for lifting. This pointed coherence is essential: at $q = 19$ an individual contracted fiber has no split witness avoiding its marker, but no consecutive-row polar line is contained in its orbit.

The resulting structural theorem has two outputs.

Theorem I.2 (Classification results and explicit gates beyond five). The following statements hold.

(a) For every prime power $q \geq 7$, the projective deep-hole directions of $\mathrm{PRS}(q-5)$ are exactly the persistent tangent and conjugate-secant families, the exceptional orbits listed for $q = 7, 8, 9, 11, 13$, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit.

(b) The displayed finite domains through $q = 32$ have the certified split-free classification stated in Section VI. Together with the structural theorem, these finite classifications give the all-field result: for every $q \geq 13$ the split-free locus consists of the persistent families and, when $q = 2^m$ with m odd, one central nucleus point. This is a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$, in particular for $q \geq 11$.

At redundancy r, the semilinear orbits in the persistent conjugate-secant family are parameterized by

$$T/T^{r-1} \quad \text{modulo inversion and Frobenius,}$$

where $T = \{z \in \mathbb{F}_{q^2}^{\times} : z^{q+1} = 1\}$ is the norm-one torus and Frobenius is coefficient Frobenius. On the tangent family the unipotent coordinate transforms as $z \mapsto z + (r-1)u$; consequently its orbit decomposition changes precisely when $\operatorname{char} \mathbb{F}_q$ divides $r-1$.

The two parts have different scopes. Both are unconditional all-field syndrome classifications, but part (b) becomes a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$. Table II places the split-free result, radius gate, deep-hole conclusion, and remaining field gap in separate columns for each redundancy.

a) Relation to previous work.: The tangent and conjugate-secant families and the coding/normal-rational curve dictionary come from [1], [2]; the covering-radius input in the high-rate range is due to Seroussi and Roth [4]. The broader arc/MDS and code-automorphism background is surveyed or established in [5], [6]. Earlier explicit-family results for projective and generalized projective Reed–Solomon codes include [7], [8], [9]; the recent extension framework of [10] further clarifies the deep-hole/MDS-extension interface. These works construct families, treat punctured or lower-redundancy regimes, or analyze the extension criterion; they do not supply the complete redundancy-five cubic-pencil exhaustion proved here. Classical apolarity, catalecticants, and binary-form rank strata are treated in [11], [12]; our use of divided powers keeps the small-characteristic contraction formulas explicit. Factorization statistics in linear families [13], normal-rational-curve nuclei [14], and the general Frobenius/monodromy semantics of splitting families [15] are inputs. The coding-theory references above identify deep syndromes with uncovered points of a normal rational curve and settle the lower-redundancy families; they do not classify the redundancy-five cubic pencils. The orbit-theoretic and factorization references supply normal forms and distribution tools, but not the coherent marked contraction used here. The nucleus results locate characteristic-dependent kernels, while the present work computes which of their coherent lifts meet the PRS splitting incidence. Accordingly, this paper's contributions are separated into the redundancy-five through seven classifications and the conditional transverse induction mechanism. Claim-by-claim novelty and dependency boundaries are recorded in the electronic ledger.

b) Organization.: Section II first gives a proof-free guide to the argument, and Section III gives the Hankel dictionary. Section IV proves Theorem I.1. Sections V and VI develop coherent polar induction and prove Theorem I.2. Section VII states the computational and formal trust boundary, and Section VIII records the exact open boundary. Companion papers will treat the fixed-level redundancy-eight and redundancy-nine problems and the logically separate ordered-Hessian and Lucas-carrier geometry.

II. Guide to the proof

This section is a proof-free guided tour. Its purpose is to install the information flow used throughout the paper before coordinates, finite exceptional tables, or formal trust boundaries intervene.

a) Syndrome to splitting system.: A syndrome direction $[f] \in \mathbb{P}(\Gamma^{r-1}E)$ determines its Hankel annihilator $W_f \subset \mathrm{Sym}^{r-2} E^\vee$. A product of $r-2$ distinct rational linear factors lies in W_f exactly when f lies in the span of the corresponding normal-rational-curve columns. Thus the geometric classification begins with split-free systems. The separate covering-radius theorem is what promotes split-free directions to code deep holes; the dictionary alone does not.

b) The redundancy-five base cover.: At redundancy five, W_f is a pencil of binary cubics. Its root incidence is a degree-three map of projective lines. The off-diagonal fiber square separates the cyclic cases, controlled by Frobenius on a deck transformation, from the S_3 case, controlled by rational points on a geometrically integral $(2,2)$ curve. This is the bottom cover used at every later level.

c) Why independent lower fibers fail.: At higher redundancy, choosing one lower witness forgets which factor was removed. If that lower witness contains the chosen factor, then multiplication restores it with multiplicity two rather than producing a squarefree form. The lost datum is therefore not auxiliary: it is the exact obstruction to lifting.

d) Coherent replacement and the fork.: Choose a prospective root λ, contract f to $\iota_\lambda f$, and retain λ as a marked forbidden root. Iteration gives an ordered polar flag. Its first-polar line has two qualitatively different behaviors. If it is transverse to the lower bad locus and collision schemes, only finitely many parameters must be removed and a point count produces an escape. If it is scheme-theoretically contained, the point-count argument says nothing: ordinary catalecticant components and components created in small characteristic require a separate level-specific classification. Section V introduces the terms persistent and modular carriers only when that distinction enters the proof.

e) Lifting.: On the transverse branch, a rational point of the lower ordered splitting cover outside the branch, diagonal, old-marker, new-marker, and self-collision divisors produces a split squarefree lower witness avoiding all markers. Multiplying by the marked factors then lifts it squarefreely to W_f. Figure 1 records this sequence and the contained/transverse fork.

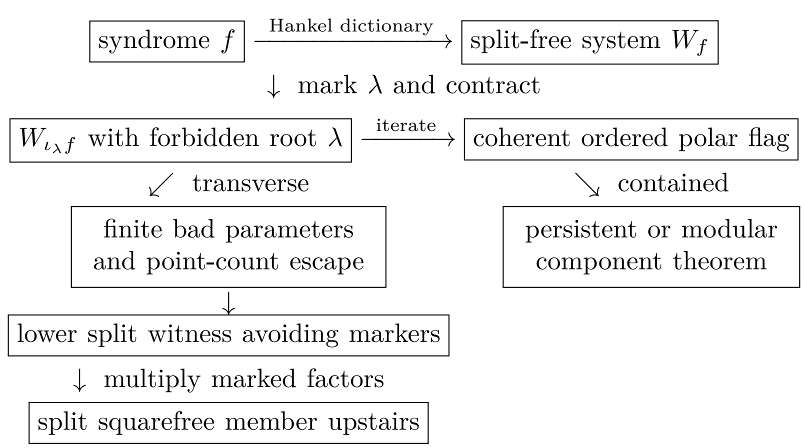

Fig. 1. Information flow of coherent polar induction. Solid arrows are the statements proved in the degrees treated here. The contained branch is not a consequence of the transverse sieve.

TABLE I
Mathematical dependency map.

Result	Printed geometric argument	Imported input
R5	Hankel dictionary; gcd strata; cyclic descent; integral $(2,2)$ fiber square and point bound	lower tangent/secant families; high-rate radius theorem; singular-curve bound
R6	one-marker polar line; ordinary and binary contained components; transverse escape	R5 lower package and radius gate
R7	two-marker contraction; contained rank-two theorem; transverse escape	R5/R6 lower packages; separate covering-radius result

TABLE II
Classification status by redundancy. "Unconditional" refers to the displayed field range. A split-free classification becomes a deep-hole classification only after the separate radius gate.

Red.	Field range	Split-free geometry	Deep holes/MDS extensions	Remaining gap
5	$q \geq 7$	Complete, unconditional	Complete, unconditional; $\rho = 4$	None
6	$q \geq 7$	Complete, unconditional	Complete, unconditional; $\rho = 5$	None
7	$q \geq 7$	Complete, unconditional	Complete for $q \geq 11$; conditional on $\rho = 6$ at $q = 7, 8, 9$	Radius at $q = 7, 8, 9$

f) Referee roadmap.: The shortest route to the headline R5 theorem is Lemma III.1 and Corollary III.2, followed by the radius gate and gcd strata in Propositions IV.1–IV.3, the cubic incidence in Proposition IV.4, the cyclic and S_3 analyses in Lemmas IV.5–IV.6, and the finite bridge in Proposition IV.7. Tables IV and V make the bounded-field output locally identifiable and numerically auditable.

The later R6 and R7 proofs use the R5 splitting cover as their bottom package. Section V may be read as the abstract mechanism linking these levels. Running the supplement is required only to reproduce the finite bridges, canonical orbit records, and regression or formal checks identified in Section VII; it is not a substitute for any printed geometric argument.

The proofs now follow these arrows in order: dictionary, redundancy-five base cover, polar construction, contained components, transverse point count, and fixed-level applications. This is a scope boundary, not a second theorem inventory.

III. The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \ldots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \ldots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. The perfect divided-power/symmetric-power pairing is characterized by coefficient extraction. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \qquad W_f = \ker H_f \subset \operatorname{Sym}^{r-2} E^\vee.$$

TABLE III
Notation used from the Hankel dictionary onward.

Symbol	Meaning
q	field size; all numerical thresholds are thresholds on prime powers
k	code dimension in $\mathrm{PRS}(k)$
$r = q + 1 - k$	redundancy
$n = r - 1$	divided-power degree of a syndrome representative
E	a two-dimensional vector space over the current base field
$f \in \Gamma^n E$	divided-power representative of a projective syndrome
$W_f \subset \operatorname{Sym}^{n-1} E^\vee$	Hankel/apolar witness system annihilated by f
g, δ, κ	genus, deletion degree, and point-bound correction
b, c	transverse bad-locus and marker-collision degrees

Lemma III.1 (Hankel criterion). Let $\lambda_1, \ldots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then

$$f \in \operatorname{span}\big(\nu(\lambda_1), \ldots, \nu(\lambda_{r-2})\big) \iff g \in W_f.$$

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\mathrm{P\Gamma L}_2(q)$.

Proof. On an affine chart, g is the characteristic polynomial of an order-$(r-2)$ recurrence. The length-r solutions are spanned by the geometric sequences $\nu(\lambda_i)$; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. □

Corollary III.2 (Split-free criterion). Assume $\rho(\mathrm{PRS}(q+1-r)) = r-1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r-2$ over $\mathbb{F}_q$.

We call the latter property split-free even when the covering-radius hypothesis has not been established. Thus every deep syndrome in the radius-$r-1$ range is split-free, while a small-field split-free census is not silently promoted to a code deep-hole classification. We call a direction shallow when W_f does contain a completely split squarefree form.

The first locus that persists under contraction is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy: since $\dim W_f = r-3$,

$$r-3 \le r-1-\deg\gcd(W_f),$$

so its degree is at most two.

A. Coding consequences and recognition

The dictionary has two immediate coding consequences. First, a split-free direction is an uncovered point of the normal rational curve and hence gives a one-column MDS extension whenever the stated covering-radius gate applies. The classification theorems therefore give exact projective deep-hole counts, not only normal forms. Second, they give recognition procedures. From a syndrome one forms H_f, computes W_f, and tests the listed geometric families: a common quadratic factor detects the persistent tangent and conjugate-secant families; the relevant contraction kernel detects a modular nucleus; and the finite exceptional fields are decided by the canonical orbit representatives printed in Table IV and recorded in the supplement.

For each fixed redundancy treated here, constructing and row-reducing the Hankel matrix and performing the listed family-membership tests costs $O(r^3)$ field operations. This count treats factorization of the displayed bounded-degree forms and access to the exceptional orbit tables as fixed-degree work. It excludes field construction, bit complexity, and the one-time generation and certification of those tables; an implementation may also charge separately for its chosen orbit-normalization routine. Thus the bound describes the online linear-algebra stage at the fixed redundancies proved here. It is not an end-to-end uniform arbitrary-r complexity claim, a general decoder, or a preprocessing bound. A direct witness search would instead inspect the projective members of the $(r-3)$-dimensional system W_f and factor degree-$(r-2)$ forms; coherent polar induction is the mechanism that replaces that ambient search in the levels proved below.

IV. Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

These are divided-power coordinates; the coefficients of a cubic in W_f are ordinary symmetric-power coordinates. Factoring the formal quartic $\sum a_i T^i U^{4-i}$ would therefore be noninvariant in characteristics two and three.

Proposition IV.1 (Radius gate). For every prime power $q \ge 7$,

$$\rho(\mathrm{PRS}(q-4)) = 4.$$

Proof. The coding/arc criterion identifies radius four with completeness of the quartic normal rational curve. Seroussi–Roth prove completeness when

$$k \ge \left\lfloor \frac{q-1}{2} \right\rfloor$$

apart from the even-field dimensions $k = 2, q-2$ [4]. Here $k = q-4$, the inequality holds for $q \ge 7$, and neither exceptional dimension occurs. Thus Corollary III.2 applies. The finite certificate independently checks the same conclusion only in its declared finite domain; it is not the all-field radius proof. □

A. Gcd strata

Proposition IV.2 (Quadratic gcd). If $\deg\gcd(W_f)=2$, then

$$W_f = Q\langle T,U\rangle.$$

If Q is split squarefree, f is shallow. If Q is irreducible, f belongs to the conjugate-secant family; if $Q=\lambda^2$, it belongs to the tangent family. Conversely every point in those families has the displayed gcd. Their deep-family sizes are

$$|\mathrm{T}| = q(q+1), \qquad |\mathrm{S}| = \frac{q(q^2-1)}{2}.$$

Proof. The equality follows by dimension. A split Q and a linear cofactor avoiding its two roots give a split squarefree cubic. An irreducible Q survives in every member, while a square Q forces a repeated root in every member. Lemma III.1, including its confluent and conjugate-root forms, identifies the corresponding syndromes with the secant and tangent spans. The sizes are the established tangent/conjugate-secant counts from the lower-redundancy classification [1], [2]. □

Proposition IV.3 (Linear gcd). Suppose $\deg\gcd(W_f)=1$. Then f is shallow for $q\geq 8$. At $q=7$ the same conclusion is part of Certificate R5.

Proof. Write $W_f=\lambda_0 V$ with V a basepoint-free quadratic pencil. If its degree-two map $\psi:\mathbb{P}^1\to\mathbb{P}^1$ is separable, its rational deck involution ι gives the graph

$$\Gamma_\iota(\mathbb{F}_q) = \{(t,\iota(t)) : t\in\mathbb{P}^1(\mathbb{F}_q)\},$$

which has exactly $q+1$ rational points. A safe union bound deletes at most two fixed graph points, at most four graph points over the branch fibers, and at most the two graph points $(\lambda_0,\iota(\lambda_0))$ and $(\iota(\lambda_0),\lambda_0)$. Thus fewer than $q+1$ graph points are deleted when $q\geq 8$. A surviving point gives a split quadratic with two distinct roots avoiding λ_0, and hence a split squarefree cubic in W_f. In characteristic two an inseparable quadratic pencil is equivalent to $\langle T^2,U^2\rangle$; the two overlapping Hankel equations then force f onto the normal rational curve, contrary to the rank-two pencil hypothesis. □

B. The trivial-gcd cubic cover

Choose a basis c_1,c_2 of a trivial-gcd pencil and put

$$X_f = \{([u:v],t) : uc_1(t)+vc_2(t)=0\}\subset\mathbb{P}^1\times\mathbb{P}^1.$$

Proposition IV.4 (Cubic incidence and off-diagonal fiber square). The curve X_f is geometrically integral of bidegree $(1,3)$ and arithmetic genus zero. Projection to the root coordinate is birational, while projection to the pencil coordinate gives a degree-three morphism

$$\phi_f:\mathbb{P}^1\longrightarrow\mathbb{P}^1.$$

If ϕ_f is separable, division of

$$c_1(t_1)c_2(t_2)-c_2(t_1)c_1(t_2)$$

by the diagonal bracket defines a residual curve $Y_f\subset\mathbb{P}^1\times\mathbb{P}^1$ of bidegree $(2,2)$ and arithmetic genus one. Its geometric components are the geometric-monodromy orbits on ordered pairs of distinct roots.

Proof. A component of X_f of bidegree $(0,e)$ would be a common factor of c_1,c_2; hence none exists. The bidegree genus formula gives genus zero. A root determines the unique pencil member containing it, so the root projection is birational. For a separable map the diagonal factor in the fiber-square determinant has multiplicity one. Removing its bidegree $(1,1)$ from the fiber square leaves bidegree $(2,2)$, whose arithmetic genus is one. The standard fiber-square/monodromy correspondence identifies its components with orbits on distinct ordered sheets. □

Lemma IV.5 (Cyclic stratum). Suppose ϕ_f is geometrically cyclic.

(i) In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when $q\equiv 2 \pmod 3$; if they are conjugate, it is deep exactly when $q\equiv 1 \pmod 3$.

(ii) In characteristic three all osculating planes meet in $n=(0:0:1:0:0)$, whose pencil is $\langle T^3,U^3\rangle$; this fixed point is deep.

(iii) The remaining wild characteristic-three pencils have the form $\langle U^3, T^3+aTU^2\rangle$. They are deep exactly when $-a$ is nonsquare, forming one orbit of size $(q^2-1)/2$.

Proof. Assume first that the characteristic is not three. Riemann–Hurwitz gives two totally ramified points t_1, t_2. The confluent Hankel criterion places f in $\mathrm{Osc}(t_1) \cap \mathrm{Osc}(t_2)$; after transporting the pair to $\{0, \infty\}$, the pencil is $\langle T^3, U^3 \rangle$ and a geometric deck generator is $t \mapsto \zeta t$.

For a rational ramification pair, Frobenius fixes the points and the deck map descends exactly when $\zeta^q = \zeta$, namely $q \equiv 1 \pmod 3$. Thus its complement is deep exactly for $q \equiv 2 \pmod 3$. The setwise stabilizer of the ordered normal form is the split dihedral group of order $2(q-1)$, so the orbit size is

$$\frac{q(q^2-1)}{2(q-1)} = \frac{q(q+1)}{2}.$$

For a conjugate pair Frobenius interchanges the fixed points and inverts the multiplier; descent is instead $q \equiv 2 \pmod 3$. The nonsplit dihedral stabilizer has order $2(q+1)$ and the orbit size is $q(q-1)/2$.

Now suppose the characteristic is three. The second Hasse derivative of

$$\nu(t) = (1, t, t^2, t^3, t^4)$$

is $(0, 0, 1, 3t, 6t^2) = e_2$, including in the reversed infinity chart. Thus every osculating plane contains $n = [e_2]$. Substitution shows that n is $\mathrm{PGL}_2(q)$-fixed and $W_n = \langle T^3, U^3 \rangle$, so it is deep.

A remaining separable cyclic cover has one wild totally ramified point. Sending it to infinity and comparing coefficients under a nontrivial translation gives

$$W_f = \langle U^3, T^3 + \alpha T^2 U + \beta T U^2 \rangle, \qquad \alpha = 0, \quad \kappa^2 = -\beta.$$

It is therefore represented by

$$f_a = e_2 - a e_4, \qquad W_{f_a} = \langle U^3, T^3 + a T U^2 \rangle.$$

The affine members are $z^3 + az + s$. Such a member has three rational roots exactly when the additive map $z \mapsto z^3 + az$ has a nonzero rational kernel, equivalently when $-a$ is a square. Hence the nonsquare parameters form the deep orbit. Its stabilizer is $t \mapsto \pm t + \beta$, of order $2q$, and orbit–stabilizer gives $(q^2-1)/2$. ☐

Lemma IV.6 (S_3 stratum). If ϕ_f has geometric monodromy S_3 and $q \geq 23$, then the pencil contains a completely split squarefree cubic.

Proof. By Proposition IV.4, components of Y_f correspond to monodromy orbits on distinct ordered roots. The S_3 action is two-transitive, so Y_f is geometrically integral and has arithmetic genus one. Aubry–Perret gives

$$\# Y_f(\mathbb{F}_q) \geq q - 2\sqrt{q}.$$

The diagonal costs

$$Y_f \cdot \Delta = (2,2) \cdot (1,1) = 4.$$

Riemann–Hurwitz gives different degree four for the separable cubic map, hence at most four branch values. A singular cubic has at most two distinct roots and at most two ordered off-diagonal pairs, so the branch deletion costs at most eight. If $q - 2\sqrt{q} > 12$, a rational off-diagonal unramified point remains. It gives two distinct rational roots of a squarefree cubic, whose third root is rational. The inequality holds for every prime power $q \geq 23$. ☐

Proposition IV.7 (Exhaustion and finite bridge). The preceding cases exhaust every redundancy-five pencil for $q \geq 23$. Certificate R5 closes exactly

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19\}.$$

It finds sporadic S_3-stratum orbits exactly at $q = 7, 8, 9, 11, 13, 17, 19$ and none at $q = 16$.

Proof. The gcd degree is zero, one, or two. Propositions IV.2 and IV.3 settle the positive-gcd cases. In the trivial-gcd case the cubic map is inseparable or separable. The inseparable case occurs only in characteristic three and the overlap equations give the all-cube nucleus pencil. A separable transitive cubic has geometric monodromy C_3 or S_3, settled by Lemmas IV.5 and IV.6. The stated finite residue, canonical representatives, stabilizers, histograms, and Frobenius links are exactly the domain of Certificate R5. ☐

TABLE IV: Complete sporadic orbit inventory at redundancy five. A representative $[a_0, \dots, a_4]$ is in the divided-power syndrome coordinates of Section III. The last column records the nontrivial Frobenius fusion; a dash means the orbit is fixed.

q	representative	orbit size	stabilizer	Frobenius
7	[0,1,0,0,3]	28	12	–
7	[0,1,0,0,2]	112	3	–
7	[0,0,1,0,3]	168	2	–
7	[0,1,0,3,2]	168	2	–
7	[1,0,0,2,1]	168	2	–
8	[0,1,1,0,2]	252	2	$\to [0, 1, 1, 0, 4]$
8	[0,1,1,0,4]	252	2	$\to [0, 1, 1, 0, 6]$
8	[0,1,1,0,6]	252	2	$\to [0, 1, 1, 0, 2]$
9	[1,0,0,1,2]	180	4	–
9	[0,1,0,4,1]	360	2	$\leftrightarrow [0, 1, 0, 4, 4]$
9	[0,1,0,4,4]	360	2	$\leftrightarrow [0, 1, 0, 4, 1]$
11	[1,0,0,1,10]	330	4	–
11	[1,0,0,1,1]	660	2	–
13	[0,1,0,0,4]	182	12	–
13	[1,0,0,4,7]	546	4	–
17	[1,0,0,1,5]	1224	4	–
19	[0,1,0,0,4]	570	12	–

The certificate additionally records the complete cubic-pencil member histogram of each representative. Repeated orbit sizes in Table IV are therefore visibly distinguished without requiring a working-directory file.

TABLE V

Independent R5 completeness totals. The first equality is the direct syndrome enumeration versus the orbit–stabilizer sum; the second decomposes the same total into the formula in Theorem I.1 and the sporadic rows above.

q	direct = orbit sum	nonsporadic	sporadic
7	$889 = 889$	245	644
8	$1116 = 1116$	360	756
9	$1391 = 1391$	491	900
11	$1848 = 1848$	858	990
13	$2080 = 2080$	1352	728
16	$2432 = 2432$	2432	0
17	$4131 = 4131$	2907	1224
19	$4541 = 4541$	3971	570

V. Coherent polar induction

A. Spaces, markers, and bad schemes

Fix a field k, a two-dimensional k-space E, and

$$S_n = \mathbb{P}(\Gamma^n E).$$

For $\lambda \in E^\vee$, define divided-power contraction by

$$\langle \iota_\lambda f, g\rangle = \langle f, \lambda g\rangle.$$

Then

$$g \in W_{\iota_\lambda f} \quad \Longleftrightarrow \quad \lambda g \in W_f. \tag{1}$$

The lower witness lifts squarefreely only when $\lambda \nmid g$. Thus the factor selected for contraction must remain as a marked forbidden root.

Definition V.1 (Carriers). A carrier is a closed locus on which the transverse splitting argument degenerates. A persistent carrier is an ordinary catalecticant component propagated by contraction; a modular carrier is an additional component created by a positive-characteristic contraction kernel.

Definition V.2 (Contraction and marker spaces). Let $C_f : E^\vee \to \Gamma^{n-1}E$ be $\lambda \mapsto \iota_\lambda f$, and put $U_f = \mathbb{P}(E^\vee) \setminus \mathbb{P}(\ker C_f)$. The first-polar rational map is

$$\Phi_f : U_f \longrightarrow S_{n-1}, \qquad [\lambda] \longmapsto [\iota_\lambda f].$$

Its image is a point or a line. For $j \geq 1$, let

$$\mathrm{Conf}_j(\mathbb{P}(E^\vee)) = \{(\lambda_1, \dots, \lambda_j) : \lambda_i \neq \lambda_\ell \text{ for } i \neq \ell\}.$$

The ordered marker space $U_{f,j}$ is the open subset on which every successive contraction is nonzero. The universal j-step polar flag is the map

$$(\lambda_1, \dots, \lambda_j) \longmapsto \big([\iota_{\lambda_1} f], \dots, [\iota_{\lambda_j} \cdots \iota_{\lambda_1} f]\big).$$

In an affine coordinate r the first map is

$$\Phi_f(r) = (a_1 - ra_0, \dots, a_n - ra_{n-1}), \qquad \Phi_f(\infty) = (a_0, \dots, a_{n-1}).$$

Example V.3 (The complete R5-to-R6 propagation at $q = 29$). Work over $\mathbb{F}_{29}$ at redundancy six and take

$$f = (6 : 1 : 0 : 0 : 5 : 1).$$

The Hankel kernel is the net of quartics

$$W_f = \ker \begin{pmatrix} 6 & 1 & 0 & 0 & 5 \\ 1 & 0 & 0 & 5 & 1 \end{pmatrix} = \langle (0,0,1,0,0), (24,1,0,1,0), (28,1,0,0,1) \rangle .$$

The displayed basis has no common factor, so the persistent test does not decide the syndrome. Choose the first marker $r = 0$. Then

$$\iota_0 f = (1 : 0 : 0 : 5 : 1).$$

This is now a redundancy-five syndrome, and its full cubic pencil is

$$W_{\iota_0 f} = \langle (0,1,0,0), (25,0,1,24) \rangle .$$

The member with pencil parameter $[9 : 28]$ is

$$h(t) = 4 + 9t - t^2 + 5t^3 \in W_{\iota_0 f},$$

because the two Hankel products are $4 + 5 \cdot 5 = 0$ and $5(-1) + 5 = 0$ in $\mathbb{F}_{29}$. Direct factorization gives

$$h(t) = 5(t-2)(t-6)(t-27).$$

None of these roots is the marked root 0. Equation (1) therefore lifts h to

$$t\,h(t) = (0, 4, 9, 28, 5) \in W_f,$$

a squarefree quartic with roots $0, 2, 6, 27$. The Hankel criterion now places f in the span of the corresponding four normal-rational-curve columns, so f is shallow. This one calculation exhibits every interface in the R5-to-R6 propagation: the upper Hankel net, the ordinary common-factor test, contraction to the complete R5 cubic pencil, selection on its splitting cover, marker avoidance, and squarefree lifting. In the general proof the R5 curve argument supplies such a pencil parameter after deleting the marked fibers; the displayed parameter $[9 : 28]$ is one complete instance of that step.

Definition V.4 (One-step lower package). Fix $f \in S_n(\mathbb{F}_q)$ with injective contraction map, so that $\Phi_f : \mathbb{P}(E^\vee) \to S_{n-1}$ is defined everywhere. A one-step lower package for f consists of:

(i) a closed bad scheme $B_{n-1} \subset S_{n-1}$ and a finite stratification $S_{n-1} \setminus B_{n-1} = \coprod_\alpha X_\alpha$;

(ii) for every $\lambda \in \mathbb{P}(E^\vee)(\mathbb{F}_q)$ with $b = \iota_\lambda f \in X_\alpha(\mathbb{F}_q)$, a specified proper geometrically integral curve $Y_{\alpha,b}$, defined over $\mathbb{F}_q$, with genus bound g_α and point bound

$$\#Y_{\alpha,b}(\mathbb{F}_q) \geq q + \kappa_\alpha - 2g_\alpha\sqrt{q};$$

(iii) a divisor $D_{\alpha,b,\lambda}$ on $Y_{\alpha,b}$, of degree at most δ_α, containing the branch, diagonal, marker, and self-collision points, such that every rational point outside $D_{\alpha,b,\lambda}$ determines an ordered rational split squarefree $g \in W_b$ with $\lambda \nmid g$;

(iv) a finite parameter scheme $Z_f \subset \mathbb{P}(E^\vee)$, of degree at most d, containing $\Phi_f^{-1}(B_{n-1})$ and every parameter at which the preceding construction is unavailable.

The index α includes any constant-field or Frobenius twist; the assertion in (iii) is the required descent statement. All degrees are geometric degrees on the displayed curve or parameter line.

The construction is summarized in Figure 2. The upper branch is the level-specific component calculation; the lower branch is the general transverse argument.

$[f] \in S_n \dashrightarrow \Phi_f(U_f) \subset S_{n-1} \longrightarrow$ lower ordered splitting cover

$\searrow$ contained   persistent/Lucas component,

$\nearrow$ transverse   delete branch, diagonal, and markers

$\downarrow$ lift by (1)

split squarefree witness in W_f.

Fig. 2. Coherent polar induction. The contained branch is a level-specific component theorem; only the transverse branch is controlled by the general point-count argument.

Theorem V.5 (Polar-flag construction). The maps of Definition V.2 commute with field extension and the natural $\mathrm{GL}(E)$ action. They include the infinity chart displayed above. If $g \in W_{\iota_{\lambda_j}\cdots\iota_{\lambda_1} f}$, then $\lambda_1 \cdots \lambda_j g \in W_f$; the lift is squarefree exactly when g is squarefree and avoids every marker λ_i.

Proof. Contraction is defined by the perfect divided-power/symmetric-power pairing, so it commutes with base change and change of basis. Repeated application of (1) gives the membership assertion. Since the λ_i are distinct on the configuration space, their product is squarefree; the product with g is squarefree precisely when g is squarefree and none of the λ_i divides it. The coordinate formula at infinity follows by evaluating C_f on the second basis covector. $\square$

Definition V.6 (One-step contained-component assertion). For a specified bad scheme B_{n-1}, collision section, and explicit closed list $\mathcal{C}_n \subset S_n$, the assertion $\mathrm{CC}(n, 1; \mathcal{C}_n)$ says that every f for which the contraction map is noninjective, $\Phi_f^* B_{n-1}$ is the whole parameter line, or the collision section vanishes identically belongs to $\mathcal{C}_n$. When the list is clear we write $\mathrm{CC}(n, 1)$.

This is a level-specific theorem obligation, not an assumption hidden inside the point-count argument. Section VI states the contained-component calculations used at redundancies six and seven. No assertion $\mathrm{CC}(n, 1)$ is made for an arbitrary degree.

Theorem V.7 (One-step polar escape). Let $f \in S_n(\mathbb{F}_q)$ have a one-step lower package as in Definition V.4. Put

$$\mathcal{H}_\kappa(g, \delta) = \left\lfloor \left(g + \sqrt{g^2 + \delta - \kappa}\right)^2 \right\rfloor + 1$$

and assume $\delta_\alpha \geq \kappa_\alpha - g_\alpha^2$ for every α. If

$$q \geq \max_\alpha \mathcal{H}_{\kappa_\alpha}(g_\alpha, \delta_\alpha) \quad \text{and} \quad q + 1 > d,$$

then W_f contains a split squarefree member.

Proof. Since $\#\mathbb{P}^1(\mathbb{F}_q) = q + 1 > d$, choose $\lambda \notin Z_f(\mathbb{F}_q)$. On the corresponding lower stratum the recorded point bound gives

$$q + \kappa_\alpha - 2g_\alpha\sqrt{q} > \delta_\alpha,$$

so there is a rational point outside all deletions. It yields a split squarefree lower member avoiding λ. Equation (1) lifts it to a split squarefree member upstairs, so the original system is shallow. $\square$

The theorem isolates only the numerical escape and one-step lifting argument. A multistep polar flag is obtained by proving and applying a new one-step package at each intermediate level; no arbitrary-j component theorem is asserted.

Remark V.8 (Point-count convention). The correction κ records the exact lower bound available for the chosen model. The singular cubic fiber square of Lemma IV.6 uses $\kappa = 0$. The normalized genus-one covers in the fixed-level applications use $\kappa = 1$. Thus the redundancy-eight triple $(g, \delta, \kappa) = (1, 30, 1)$ gives the real cutoff $(1 + \sqrt{30})^2 < 42$, whose first prime power is 43; the redundancy-nine triple $(1, 36, 1)$ gives the strict cutoff $q > 49$, whose first prime power is 53.

TABLE VI
Normalized point-count thresholds used in this paper.

Application	g	δ	κ	integer lower bound	first prime power
R5 cubic fiber square	1	12	0	$q \geq 22$	23
R6 marked cover	1	18	0	$q \geq 29$	29
R7 marked cover	1	24	0	$q \geq 37$	37

B. Contained-component calculations

The ordinary rank-two carrier admits a uniform contained-line calculation. This does not classify other lower bad schemes, but it closes the persistent part of every level-specific component gate.

Proposition V.9 (Contained rank-two polar lines). Let $n \geq 5$, let $f \in \mathbb{P}(\Gamma^n E)$, and let

$$A = C_f^{(2)} : \operatorname{Sym}^{n-2} E^\vee \longrightarrow \Gamma^2 E$$

be the three-row catalecticant. Assume the contraction map $E^\vee \to \Gamma^{n-1} E$ is injective, so the first-polar map is defined on all of $\mathbb{P}(E^\vee)$. This is exactly the complement of the rank-one boundary: after a change of coordinates, a nonzero kernel vector is ∂_X, and $\partial_X f = 0$ forces f to be a scalar multiple of $Y^{[n]}$. Such a point lies on the normal rational curve and is already excluded from the split-free locus. Then

$$\Phi_f(\mathbb{P}(E^\vee)) \subset \{b : \operatorname{rank} C_b^{(2)} \leq 2\} \quad \Longleftrightarrow \quad \operatorname{rank} A \leq 2$$

scheme-theoretically and in every characteristic. If $\operatorname{rank} A = 3$, the rank-drop parameters form a subscheme of $\mathbb{P}(E^\vee)$ of degree at most three.

Proof. For a nonzero $\lambda \in E^\vee$, multiplication identifies $\operatorname{Sym}^{n-3} E^\vee$ with the hyperplane

$$H_\lambda = \lambda \operatorname{Sym}^{n-3} E^\vee \subset \operatorname{Sym}^{n-2} E^\vee,$$

and functoriality gives

$$C^{(2)}_{\iota_\lambda f} = A|_{H_\lambda}.$$

The forward implication is immediate. Conversely, suppose $\operatorname{rank} A = 3$ and put $K = \ker A$, so $\dim K = n - 4$. If the lower rank is at most two, rank–nullity on the $(n-2)$-dimensional space H_λ gives

$$\dim(K \cap H_\lambda) \geq (n-2) - 2 = n - 4 = \dim K.$$

Thus $K \subset H_\lambda$. Containment for every geometric $[\lambda]$ would imply

$$K \subset \bigcap_{[\lambda] \in \mathbb{P}(E_{\bar{k}}^\vee)} \lambda \operatorname{Sym}^{n-3} E_{\bar{k}}^\vee = 0,$$

because a nonzero binary form cannot be divisible by every linear form. This contradicts $\dim K = n - 4 > 0$.

The maximal minors of $C^{(2)}_{\iota_\lambda f}$ are homogeneous cubics in λ. Vanishing at every geometric point is therefore identical vanishing, proving the scheme-theoretic assertion. In the noncontained case at least one cubic is nonzero, and the common rank-drop scheme lies in its degree-three zero divisor. □

Thus Proposition V.9 covers the entire non-rank-boundary locus on which the first-polar line is defined; the excluded noninjective locus is the rank-one normal-rational-curve locus just identified. After rank one and rational split secants are removed as shallow, the upper rank-two locus consists of the persistent tangent and conjugate-secant carriers. On a persistent fiber the nonsplit torus acts by $z \mapsto z^n$, giving T/T^n modulo inversion and Frobenius, while the tangent unipotent cocycle is $z \mapsto z + nu$.

Modular carriers are equally explicit. If $\mathcal{N}$ is a nucleus in the lower divided-power module, the complete space whose first-polar line lies in $\mathcal{N}$ is

$$\mathcal{M}_n(\mathcal{N}) = \mathbb{P} \ker \left(\Gamma^n E \longrightarrow \operatorname{Hom}(E^\vee, \Gamma^{n-1} E / \widetilde{\mathcal{N}}) \right).$$

Lucas' theorem computes the nucleus coordinates, and this displayed kernel computes every coherent lift without an ambient syndrome census.

Remark V.10 (Scope of the contained calculations). The modular contraction kernel is an exact finite linear-algebra construction. Proposition V.9 supplies the ordinary persistent component uniformly, but other lower bad schemes and marker collisions still require level-specific calculations. Neither the proposition nor Theorem V.7 classifies those additional components.

VI. Redundancies six and seven: complete and gated results

A. Redundancy six

For

$$f = (a_0 : \cdots : a_5) \in \mathbb{P}(\Gamma^5 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \end{pmatrix},$$

the system W_f is a net of quartics off the quintic normal rational curve. It is split-free exactly when it has no completely split squarefree quartic.

Proposition VI.1 (Persistent net and orbit law). If $\gcd(W_f) = Q$ has degree two, then

$$W_f = Q \operatorname{Sym}^2 E^\vee.$$

The split-free persistent locus has size $q(q+1)^2/2$. On the tangent fiber the unipotent coordinate is $x \mapsto x+5u$; on the irreducible quadratic fiber the norm-one torus acts through $z \mapsto z^5$, with inversion and coefficient Frobenius acting on T/T^5.

Proof. For fixed Q, the syndromes annihilating $Q \operatorname{Sym}^3 E^\vee$ form a projective line. If Q is irreducible, all $q+1$ points remain; if $Q = \lambda^2$, one endpoint has dependent Hankel rows, leaving q points. Hence

$$q(q+1) + \frac{q(q-1)}{2}(q+1) = \frac{q(q+1)^2}{2}.$$

Normalize a square to U^2. The nondegenerate fiber is $[e_4 + xe_5]$, and direct degree-five substitution gives $x \mapsto x+5u$. For an irreducible Q, its stabilizer identifies the $q+1$-point fiber with $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ and acts by fifth powers; the normalizer inverts z, while coefficient Frobenius multiplies T/T^5 by the characteristic. This proves the stated orbit law and the characteristic-five tangent splitting. □

Lemma VI.2 (Exact linear gcd is shallow). Let $q \geq 7$. If $\gcd(W_f)$ has degree one, then W_f contains a split squarefree quartic.

Proof. Write $W_f = LV$, where L is rational and $V \subset \operatorname{Sym}^3 E^\vee$ has vector dimension three. It suffices to find in V a product of three distinct rational factors, none equal to L. Move the root of L to infinity and let ℓ span $V^\perp$. On the cubic with distinct finite roots x, y, z, write

$$\ell\big((T-xU)(T-yU)(T-zU)\big) = c_0 + c_1(x+y+z) + c_2(xy+xz+yz) + c_3xyz =: P(x,y,z).$$

Suppose that this value is nonzero for every distinct triple. For fixed distinct x, y, write $P = A(x,y) + zB(x,y)$, where

$$B(x,y) = c_1 + c_2(x+y) + c_3xy.$$

If $B(x,y) \neq 0$, the unique zero in z must be x or y. Consequently

$$B(x,y)P(x,y,x)P(x,y,y) = 0$$

for every $x \neq y$; it also vanishes when $B(x,y) = 0$. This polynomial has degree at most four in each variable. Since $q-1 > 4$, first varying $y \neq x$ and then x shows that it is the zero polynomial. The polynomial ring is a domain. Neither $P(x,y,x)$ nor $P(x,y,y)$ vanishes identically unless $\ell = 0$, as comparison of the coefficients of $1, x, y, x^2, xy, x^2y$ shows. Hence $B = 0$, so $c_1 = c_2 = c_3 = 0$.

Thus V is the hyperplane of cubics vanishing at infinity, and every member of V is divisible by L. This would make $\gcd(W_f)$ divisible by L^2, contrary to its exact degree. Therefore a required cubic exists, and multiplying it by L gives a split squarefree member of W_f. □

Proposition VI.3 (First-polar line and secant degree). For $r \in \mathbb{F}_q$ put

$$b(r) = (a_1 - ra_0, \ldots, a_5 - ra_4), \qquad b(\infty) = (a_0, \ldots, a_4).$$

Then

$$(T - rU)g \in W_f \quad \Longleftrightarrow \quad g \in W_{b(r)}.$$

If W_f has trivial gcd, its polar line meets the lower secant cubic in degree at most three.

Proof. The first equivalence follows by expanding the two Hankel equations of $(T - rU)g$. Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_4 & a_5 \end{pmatrix}.$$

The lower catalecticant equals

$$C_f \begin{pmatrix} -r & 0 & 0 \\ 1 & -r & 0 \\ 0 & 1 & -r \\ 0 & 0 & 1 \end{pmatrix}.$$

Cauchy–Binet gives the secant determinant

$$D_f(r) = -\Delta_{012}r^3 + \Delta_{013}r^2 - \Delta_{023}r + \Delta_{123}.$$

It vanishes identically exactly when $\operatorname{rank} C_f \le 2$, equivalently when W_f has quadratic gcd. In the trivial-gcd case it therefore has at most three zeros. □

Proposition VI.4 (Cyclic and collision degrees). For a trivial-gcd net:

(i) in characteristic different from two and three, the lower tame cyclic carrier is a binomially rescaled Veronese surface of degree four and contains no polar line;
(ii) the pointed ramification condition has degree at most six and does not vanish identically;
(iii) an S_3 lower pencil with the marked root deleted has total deletion degree at most eighteen.

Proof. In syndrome coordinates the tame cyclic surface is

$$\left\{\operatorname{diag}(1,4,6,4,1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\operatorname{Sym}^2 E^\vee)\right\}.$$

When 2 and 3 are invertible this is the Veronese surface, of degree four, and contains no line. Characteristic three has a different wild closure, treated in Proposition VI.5. For a basis F_0, F_1, F_2 of the $g^2_4 = W_f$, pointed collision is the rank-one condition on the value and derivative rows. Equivalently, all three binary Wronskians

$$M_{ij} = F_i\, dF_j - F_j\, dF_i \qquad (0 \le i < j \le 2)$$

vanish at the marker. Each M_{ij} is a binary form of degree six: on an affine chart the apparent degree-seven leading terms cancel, and homogenization includes the point at infinity.

At least one M_{ij} is nonzero. Otherwise the morphism $\mathbb{P}^1 \to \mathbb{P}^2$ defined by the base-point-free net W_f would have zero differential and hence factor through absolute Frobenius. In characteristic three this is impossible because its pullback line bundle has degree four, not divisible by three. In characteristic two the factorization would identify W_f, after a change of coordinates, with the full pure-square space $\langle T^4, T^2U^2, U^4\rangle$. Its annihilator is $\langle e_1^\vee, e_3^\vee\rangle$. For the two consecutive Hankel rows this would require simultaneously

$$a_0 = a_2 = a_4 = 0, \qquad a_1 = a_3 = a_5 = 0,$$

contrary to $f \ne 0$. Thus the collision scheme is contained in the zero divisor of one nonzero M_{ij}, and has degree at most six in every characteristic.

Finally the R5 off-diagonal curve has the original deletion degree twelve. The unique cubic through a specified marker contributes at most six ordered pairs, giving $\delta = 18$ and

$$q - 2\sqrt{q} > 18.$$

Its first prime-power solution is 29. □

Proposition VI.5 (Modular contained components). In characteristic three the lower wild carrier is the degree-four cone

$$\operatorname{Join}(e_2, C_4),$$

and no polar line of a trivial-gcd net is contained in it. In characteristic two the cyclic carrier is the plane

$$\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle,$$

whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle.$$

Proof. Every line on the wild cone is a ruling. By equivariance it suffices to compare the two consecutive rows on one ruling; in the frame $\langle e_2, e_4\rangle$ the conditions

$$(a_0, \ldots, a_4),\ (a_1, \ldots, a_5) \in \langle e_2, e_4\rangle$$

force the overlapping coordinates successively to vanish. Undoing the frame gives $f = \mu\nu(t)$, including $f = \mu e_5$ at infinity. These are fixed-factor/rank-boundary syndromes, so no trivial-gcd polar line is contained in the wild cone.

In characteristic two, direct substitution gives

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0],$$

whose closure is Π_{cyc}. The consecutive-row overlap is now literal: the first row lies in the plane exactly when $a_0 = a_4 = 0$, and the second exactly when $a_1 = a_5 = 0$. Hence a polar line lies in the plane exactly when

$$a_0 = a_1 = a_4 = a_5 = 0,$$

namely when $f \in \mathbb{P}\langle e_2, e_3\rangle$. This proves both existence and uniqueness of the contained binary component. □

Proposition VI.6 (Binary nucleus arithmetic). The scheme $\mathcal{N}_3(\mathbb{F}_q)$ is one projective-semilinear orbit of $q+1$ points. It is split-free over $q = 2^m$ exactly when m is odd.

Proof. For the representative e_3,

$$W_{e_3} = \langle 1, t, t^4 \rangle.$$

The root differences of $A + Bt + Ct^4$ lie in the kernel of $u \mapsto u^4 + (B/C)u$. Four distinct rational roots exist exactly when all roots of $u^3 = B/C$ are rational, equivalently when $3 \mid q-1$, which for $q = 2^m$ means m is even. The stabilizer of e_3 is the Borel of order $q(q-1)$, so the orbit has $q+1$ points; its lower-unipotent transforms and endpoint are precisely $\mathcal{N}_3(\mathbb{F}_q)$. □

Proposition VI.7 (R6 one-step exhaustion). Assume one of

$$\operatorname{char}\mathbb{F}_q \geq 5,\ q \geq 29; \qquad \operatorname{char}\mathbb{F}_q = 3,\ q \geq 81; \qquad \operatorname{char}\mathbb{F}_q = 2,\ q \geq 32.$$

Then every split-free redundancy-six syndrome lies in the persistent quadratic-gcd locus or, when $q = 2^m$ with odd m, in $\mathcal{N}_3(\mathbb{F}_q)$.

Proof. The contraction map is noninjective only on the normal rational curve, which is shallow, by Proposition V.9. The common-factor trichotomy is exhaustive. A quadratic gcd gives the persistent tangent and conjugate-secant loci, together with the shallow rational secant; an exact linear gcd is shallow by Lemma VI.2. It remains to treat a trivial-gcd net.

Its first-polar line is not contained in the lower secant cubic, by Proposition VI.3. The pullback of that cubic has degree at most three. The tame cyclic surface has degree four and contains no polar line. In characteristic three its wild replacement has the same degree and no contained trivial-gcd polar line; in characteristic two the replacement is a plane, whose only contained component is $\mathcal{N}_3$, by Proposition VI.5. Finally, Proposition VI.4 gives a nonzero pointed-collision divisor of degree at most six. Outside the contained binary component, the excluded parameter scheme therefore has degree

$$d \leq 3 + 4 + 6 = 13.$$

For every remaining marker, the lower cubic pencil has exact linear gcd or geometric monodromy S_3. In the first case the graph argument of Proposition IV.3 is pointed as follows: besides its eight original deletions, remove the at most two graph points whose split quadratic contains the new marker. Since $q + 1 > 10$, a split squarefree cubic avoiding that marker remains. In the S_3 case, Lemma IV.6 supplies the geometrically integral off-diagonal curve. Its original diagonal and branch deletions have degree at most twelve, and deleting the unique cubic through the marker removes at most six ordered pairs. Thus

$$(g, \delta, \kappa) = (1, 18, 0), \qquad q - 2\sqrt{q} > 18$$

for every $q \geq 29$. These data form the one-step package of Definition V.4, with $d = 13$, so Theorem V.7 produces a split squarefree quartic.

The displayed characteristic ranges are exactly the fields at or above this point-count threshold after the bounded fields in the theorem below: the next characteristic-three field is 81, and the next characteristic-two field is 32. On $\mathcal{N}_3$, Proposition VI.6 gives the asserted odd-degree split-free orbit and proves shallowness in even extension degree. □

The covering-radius gate is complete: Seroussi–Roth gives $\rho(\mathrm{PRS}(q-5)) = 5$ for $q \geq 9$, while Certificate R6 checks it directly at $q = 7, 8$.

Theorem VI.8 (Redundancy six). For every prime power $q \geq 7$, the deep syndromes of $\mathrm{PRS}(q-5)$ are the persistent stratum of size $q(q+1)^2/2$, together with:

(i) respectively $18, 11, 4, 2, 1$ exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;
(ii) one characteristic-two nucleus orbit of size $q+1$ when $q = 2^m$ and odd $m \geq 5$.

There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five. At $q = 7, 8, 9, 11, 13$, the exceptional direction totals are

$$5152, \quad 4713, \quad 1800, \quad 792, \quad 546,$$

and the corresponding total numbers of deep syndrome directions are

$$5376, \quad 5037, \quad 2250, \quad 1584, \quad 1820.$$

Proof. Proposition VI.7 gives the complete conceptual exhaustion in characteristic at least five from $q \geq 29$, characteristic three from $q = 81$, and characteristic two from $q = 32$. Certificate R6 exhausts

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27\},$$

finds exceptional projective-orbit counts $18, 11, 4, 2, 1$ only at the five displayed fields, and no further orbit at the remaining fields. Together with the separate radius gate, this covers every prime power $q \geq 7$. $\square$

The listed sources are disjoint. The persistent locus has quadratic gcd, every finite exceptional row has trivial gcd, and the separate nucleus term occurs only for binary fields beyond the finite table.

The five small tables are exhaustive computations. Their semilinear orbit counts are $18, 5, 2, 2, 1$; collision energy plus, in odd characteristic, the quintic root-divisor type separates the classes up to precisely Frobenius fusion. Table VII prints the canonical representatives, stabilizers, orbit sizes, and fusion lengths.

TABLE VII: Exceptional redundancy-six classes. Each row is one projective-semilinear class. The columns give a canonical syndrome representative, the size and stabilizer order of each constituent $\mathrm{PGL}_2(q)$-orbit, the number ϕ of such orbits fused by coefficient Frobenius, the collision energy E, and the binary-quintic factor type τ_5 in odd characteristic.

q	representative	$\lvert\mathcal{O}\rvert$	$\lvert G_f\rvert$	ϕ	E	τ_5
7	$[0:0:0:1:0:1]$	168	2	1	7	1^3+2
7	$[0:0:1:0:3:1]$	336	1	1	19	1^2+1+2
7	$[0:0:1:0:3:2]$	336	1	1	15	1^2+3
7	$[0:1:0:0:0:1]$	168	2	1	10	$1+2+2$
7	$[0:1:0:0:2:0]$	112	3	1	33	$1+1+3$
7	$[0:1:0:1:1:2]$	336	1	1	9	$1+4$
7	$[0:1:0:1:1:5]$	336	1	1	9	$1+1+3$
7	$[0:1:0:1:3:0]$	336	1	1	21	$1+1+1+2$
7	$[0:1:0:1:3:3]$	336	1	1	7	$1+1+3$
7	$[0:1:0:1:3:6]$	336	1	1	14	$1+1+3$
7	$[0:1:0:3:0:5]$	168	2	1	11	$1+4$
7	$[0:1:0:3:0:6]$	168	2	1	20	$1+4$
7	$[0:1:0:3:2:1]$	336	1	1	13	$1+1+3$
7	$[1:0:0:0:1:2]$	336	1	1	7	$1+4$
7	$[1:0:0:0:1:3]$	336	1	1	23	5
7	$[1:0:0:3:1:3]$	336	1	1	13	$1+4$
7	$[1:0:1:0:5:2]$	336	1	1	8	$1+4$
7	$[1:0:1:0:6:2]$	336	1	1	8	5
8	$[0:0:0:1:0:0]$	9	56	1	0	–
8	$[0:1:1:0:2:1]$	504	1	3	20	–
8	$[0:1:1:0:2:5]$	504	1	3	19	–
8	$[1:0:0:1:3:6]$	504	1	3	14	–
8	$[1:0:1:3:5:2]$	168	3	1	54	–
9	$[0:1:0:0:0:4]$	180	4	2	28	$1+4$
9	$[0:1:0:1:4:2]$	720	1	2	29	$1+4$
11	$[0:0:0:1:0:0]$	132	10	1	60	1^3+1^2
11	$[0:1:0:2:0:10]$	660	2	1	30	$1+4$
13	$[0:1:0:0:0:2]$	546	4	1	60	$1+4$

For $q = 8$, integers are binary polynomial-basis codes with $\alpha^3 = \alpha + 1$; for $q = 9$, they are ternary polynomial-basis codes with $\alpha^2 = -1$. Thus $\sum \phi = (18, 11, 4, 2, 1)$ and $\sum \phi\lvert\mathcal{O}\rvert = (5152, 4713, 1800, 792, 546)$, in the field order $7, 8, 9, 11, 13$.

B. Redundancy seven

For

$$f = (a_0 : \ldots : a_6) \in \mathbb{P}(\Gamma^6 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 & a_5 \\ a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \end{pmatrix},$$

the system W_f is a four-dimensional web of quintics. For an affine marker r the contractions are

$$b(r) = (a_1 - ra_0, \ldots, a_6 - ra_5).$$

Coefficient collection gives

$$(t-r)g \in W_f \quad \Longleftrightarrow \quad g \in W_{b(r)};$$

the lift is squarefree exactly when $g(r) \neq 0$.

Proposition VI.9 (Two-marker lower cover). Let x be a forbidden root on a trivial-gcd quartic net. On the generic S_3 bottom fiber, the diagonal and branch divisors and the two marked-root fibers have total degree at most

$$12 + 6 + 6 = 24.$$

Hence $q - 2\sqrt{q} > 24$, and in particular every prime power $q \geq 37$, produces a split lower quartic avoiding x.

Proof. The R5 off-diagonal curve is geometrically integral of arithmetic genus one and has base deletion degree twelve. Each marker belongs to a unique cubic fiber, which contains at most six ordered pairs. Delete those two fibers and apply the $\kappa = 0$ bound. $\square$

Proposition VI.10 (Exact-gcd-one avoidance). Suppose the lower net is $\ell_x U$, where $U = \ker \lambda \subset \operatorname{Sym}^3 E^\vee$ has dimension three and $x \neq r$. For $q \geq 16$ it contains a split cubic avoiding both x and r.

Proof. Put $S = \mathbb{P}^1(\mathbb{F}_q) \setminus \{x, r\}$. For ordered distinct $u, v \in S$, the map $w \mapsto \lambda(\ell_u \ell_v \ell_w)$ is linear. If it vanishes identically, choose any remaining w; otherwise it has one rational zero. The equations $w = x, r, u, v$ have respective bidegrees $(1,1), (1,1), (2,1), (1,2)$ in (u, v). None is identically zero: the first two would add a common factor, and the last two are excluded because double-root cubics span $\operatorname{Sym}^3 E^\vee$. A nonzero bidegree-$(a,b)$ divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ has at most $(a+b)(q+1)$ rational points. Thus the bad pairs number at most $10(q+1)$, whereas $(q-1)(q-2) > 10(q+1)$ for $q \geq 16$. $\square$

Proposition VI.11 (Marked self-collision). After fixed factors are removed, the self-collision divisor of the moving g^3_5 has degree at most eight.

Proof. It is the ramification divisor of a separable g^3_5, whose degree is $(3+1)(5-3) = 8$. If the series were inseparable, it would factor through Frobenius and lie in the p-th-power subspace of quintics; that subspace has dimension at most three, smaller than the four-dimensional Hankel series. $\square$

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3, with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is deep for odd m; for even m, the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Proposition VI.12 (Central binary lift). The complete inverse image of the R6 nucleus line under consecutive sextic contraction is $[e_3]$. It is split-free over $\mathbb{F}_{2^m}$ exactly when m is odd.

Proof. Putting both consecutive rows in $\mathbb{P}\langle e_2, e_3 \rangle$ gives $a_0 = a_1 = a_2 = a_4 = a_5 = a_6 = 0$. Thus only e_3 remains, with $W_{e_3} = \langle 1, t, t^4, t^5 \rangle$. For odd m, contraction at a root of a hypothetical split quintic would give a split quartic on the R6 nucleus line, contradicting Proposition VI.6. For even m, $\mathbb{F}_4 \subset \mathbb{F}_q$ and the homogenization of $t^4 + t$ has its four affine $\mathbb{F}_4$ roots and the simple root at infinity. $\square$

Corollary VI.13 (Degree-six contained component). The assertion $\mathrm{CC}(6,1)$ for the lower rank-two carrier holds in every characteristic. Its contained sextics have upper catalecticant rank at most two; after the shallow rank-one and rational secant loci are removed, they are exactly the persistent tangent and conjugate-secant families.

Proof. Apply Proposition V.9 with $n = 6$. The three rational types of the quadratic recurrence are split squarefree, repeated rational, and irreducible. They give respectively the shallow rational secant, tangent, and conjugate-secant loci. The remaining lower binary nucleus line has the separate complete lift computed in Proposition VI.12. $\square$

Theorem VI.14 (Redundancy-seven classification). Certificate R7 unconditionally classifies every prime power $q \leq 32$ in its displayed domain. For every prime power $q \geq 13$, the split-free syndromes at redundancy seven are the persistent stratum, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles are:

7	$56,\ 84^5,\ 112^2,\ 168^{45},\ 336^{141}$
8	$63,\ 72,\ 84^3,\ 168^4,\ 252^{24},\ 504^{86}$
9	$180^3,\ 240^6,\ 360^{18},\ 720^{27}$
11	$264^2,\ 440,\ 660^2.$

This is the complete all-field split-free classification. Since Seroussi–Roth gives $\rho(\mathrm{PRS}(q-6)) = 6$ for every prime power $q \geq 11$, it is also the complete deep-hole classification in that range. At $q = 7, 8, 9$ the table is not promoted to a code deep-hole classification without a separate covering-radius calculation. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

Proof. For $q \geq 37$, Propositions VI.9–VI.11 give the transverse witness after at most three catalecticant intersections, eight self-collisions, and, in characteristic two, one intersection with the nucleus line. Corollary VI.13 leaves exactly the persistent locus and Proposition VI.12. Certificate R7 covers

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}$$

by an independent five-secant replay. The radius theorem promotes the split-free list only from $q \geq 11$. □

For the four finite rows Certificate R7 contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records.

The $q = 19$ lower net $\langle 1, t^3, t^4 \rangle$ has six split members, all containing the marked point at infinity, yet no coherent sextic polar line lies in its orbit. This is the concrete reason the R7 proof must apply Theorem V.7 twice with the old marker retained, rather than classify only a set of bad lower fibers.

TABLE VIII
Field-range closure ledger. "Certificate" means a complete finite classification with canonical representatives, not merely an orbit count.

Result	prime powers	closure mechanism
R5	$7, 8, 9, 11, 13, 16, 17, 19$	direct census and independent replay
	$q \geq 23$	cubic-cover geometry, Lemma IV.6
R6	$7, 8, 9, 11, 13, 16$	direct syndrome/radius census
	$17, 19, 23, 25, 27$	finite structural bridge certificate
	$q \geq 29$	marked-cover point bound and contained components
R7	$7, 8, 9, 11$	direct split-free census
	$13, 16, 17, 19, 23, 25, 27, 29, 31, 32$	coherent-polar finite bridge certificate
	$q \geq 37$	marked-cover point bound and contained components

The R7 rows classify split-free syndromes. Conversion to code deep-hole directions uses the separate covering-radius gate and is made only for $q \geq 11$. The public certificate schema and exact replay boundaries are given in the supplement; the table does not promote orbit-size profiles to identifiers.

VII. Verification and formal boundary

The conceptual theorems above are accompanied by deterministic evidence bundles. Table IX records which claims use exhaustive computation. Each public label resolves, through the supplement schema, to its generator, certificate, replay where stated, and SHA-256 manifest. Table I gives the corresponding claim-level mathematical dependencies.

TABLE IX: Public verification map. Exact filenames, commands, hashes, domains, and stop conditions belong to the electronic supplement.

Public label	Verified domain	Replay and trust boundary
Certificate R5	nineteen finite fields	independent pointwise and orbitwise replay; canonical representatives and exhaustion records
Certificate R6	direct scan through $q = 16$; finite bridge below 29	independent syndrome replay and separate radius gate
Certificate R6-NF	small exceptional normal forms	deterministic normal-form reconstruction; not a second implementation
Certificate R7	$q = 7, 8, 9, 11$ and finite bridge below 37	independent five-secant/orbit checks; split-free and radius flags remain separate

Table VII is a deterministic projection of the hash-pinned public R6 and R6-NF records. The top-level verifier regenerates it and checks byte-for-byte equality with the included source.

The public labels are defined by supplement/CERTIFICATE-SCHEMA.md. The paper-local bundle, literal replay commands, toolchain locks, hashes, byte counts, and working directories are recorded in supplement/REPRODUCING.md, supplement/EVIDENCE-MANIFEST.json, and supplement/RELEASE-MANIFEST.md. The single entry point python3 supplement/verify.py checks the bundle manifest, classification records and their expected hashes, and the local PDF and supplement-artifact rows printed in the release manifest; its –replay option runs the paper-local Python replays.

The Lean development has one paper-facing aggregate import gate. It checks the common Hankel and coding interfaces; redundancy-five Hankel algebra, projective scaling, family/count arithmetic, and transcribed finite-table arithmetic; coherent polar contraction and the redundancy-six/seven conditional synthesis interfaces.

The formal development does not prove the concrete projective Hankel dictionaries, geometric component classifications, Hasse–Weil existence after deletion, covering-radius theorems, genuine $\mathrm{PGL}_2/\mathrm{P\Gamma L}_2$ actions, or the semantics and exhaustion of the external finite records. These remain theorem hypotheses or structure fields. The aggregate axiom audit reports only propext, Classical.choice, and Quot.sound; there is no project-local axiom, generated evaluator, native decision procedure, or external oracle in the imported closure. The exact declaration for every manuscript theorem, its missing formal content, the aggregate gate, and the reproduction command are listed in supplement/LEAN-STATEMENTS.md.

Data, code, and disclosure

The electronic supplement accompanying this manuscript contains the source, canonical classification representatives and stabilizers, certificate schemas, generators, independent replays where available, expected hashes, toolchain locks, and a top-level verification command. Section VII assigns each computational or formal claim its exact trust route.

VIII. Scope and open problems

The first levels and the contained/transverse fork point to one stable picture. Their point-count thresholds are compatible with a quadratic uniform scale in the redundancy, while every surviving infinite family comes from one of the two component mechanisms isolated by the polar flag.

Conjecture VIII.1 (Stable polar classification). There is an absolute constant C such that, at redundancy r, every split-free projective syndrome over $\mathbb{F}_q$ with $q \geq Cr^2$ lies either in a persistent rank-two carrier or in a nonshallow component of the level-r Lucas contraction kernel. Equivalently, transverse arithmetic exceptions disappear above a quadratic field threshold. In every characteristic for which that Lucas kernel has no nonshallow component, the persistent tangent and conjugate-secant families are the complete high-field list.

The Lucas clause is forced by the binary nucleus families at redundancies six and seven, which persist over infinitely many fields. The conjecture predicts that rank-two persistence and digit-controlled contraction kernels exhaust the stable locus, with the coherent polar flag eliminating everything transverse.

The present results establish the first levels of this picture but do not constitute an arbitrary-redundancy classification.

1) Redundancy five is complete for every $q \geq 7$.
2) Redundancy six is a complete all-field deep-hole classification. Redundancy seven has a complete all-field split-free classification; its $q = 7, 8, 9$ covering-radius gate remains separate.
3) Each use of one-step polar escape requires a new level-specific monodromy/component calculation. A field census cannot replace this input.

Thus the one-step theorem does not give an arbitrary-redundancy classification, and higher-level component and modular-carrier strata remain open here. In particular, the paper does not settle the general Reed–Solomon deep-hole conjecture.

Two companion papers will take up the branches deliberately left out here. The first develops the fixed-level redundancy-eight and redundancy-nine problems. It will state no classification until the homogeneous component, marker-collision, and modular-carrier exhaustions are complete; at redundancy nine this also includes the binary-quartic slices and characteristic-five carrier. The second treats the logically separate ordered-Hessian, Lucas-carrier, and distinguished e_7 geometry. These forthcoming analyses are not inputs to any theorem in the present paper.

The structural conclusion is nevertheless stable. Beyond redundancy four, deepness is no longer controlled by one invariant or by isolated bad fibers. The persistent geometry is a coherent rank-two polar flag; modular geometry appears through characteristic-specific contraction kernels; and the remaining arithmetic is the Frobenius action on a normalized ordered-root cover. Redundancy five is the first exceptional-cover case, while redundancies six and seven show how that cover is propagated or contained.

Appendix

Acknowledgment

OpenAI Codex was used under the author's direction throughout the project for proof exploration, symbolic and finite computation, code generation, and drafting assistance. Drafting assistance affected prose throughout the manuscript and electronic supplement. The author checked the resulting arguments and computations and assumes responsibility for every claim in the manuscript. The system is not an author.

References

[1] K. Kaipa, “Deep holes and MDS extensions of Reed–Solomon codes,” IEEE Transactions on Information Theory, vol. 63, no. 8, pp. 4940–4948, Aug. 2017.
[2] J. Zhang, D. Wan, and K. Kaipa, “Deep holes of projective Reed–Solomon codes,” IEEE Transactions on Information Theory, vol. 66, no. 4, pp. 2392–2401, 2020.
[3] Y. Aubry and M. Perret, “Coverings of singular curves over finite fields,” Manuscripta Mathematica, vol. 88, pp. 467–478, 1995.
[4] G. Seroussi and R. M. Roth, “On MDS extensions of generalized Reed–Solomon codes,” IEEE Transactions on Information Theory, vol. 32, no. 3, pp. 349–354, 1986.
[5] S. Ball and M. Lavrauw, “Arcs in finite projective spaces,” EMS Surveys in Mathematical Sciences, vol. 6, no. 1–2, pp. 133–172, 2019.
[6] A. Dür, “The automorphism groups of Reed–Solomon codes,” Journal of Combinatorial Theory, Series A, vol. 44, no. 1, pp. 69–82, 1987.
[7] J. Zhang and D. Wan, “Explicit deep holes of Reed–Solomon codes,” https://arxiv.org/abs/1711.02292, 2017, version 1, 7 November 2017.
[8] X. Xu, S. Hong, and Y. Xu, “On deep holes of generalized projective Reed–Solomon codes,” https://arxiv.org/abs/1705.07823, 2017, version 1, 22 May 2017.
[9] X. Xu, “On deep holes of projective Reed–Solomon codes over finite fields with even characteristic,” Wuhan University Journal of Natural Sciences, vol. 28, no. 1, pp. 15–19, 2023.
[10] Y. Wu, C. Ding, and T. Chen, “Extended codes and deep holes of MDS codes,” https://arxiv.org/abs/2312.05534, 2023, version 1, 9 December 2023.
[11] A. Iarrobino and V. Kanev, Power Sums, Gorenstein Algebras, and Determinantal Loci, ser. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 1999, vol. 1721.
[12] G. Comas and M. Seiguer, “On the rank of a binary form,” Foundations of Computational Mathematics, vol. 11, pp. 65–78, 2011.
[13] E. Cesaratto, G. Matera, and M. Pérez, “The distribution of factorization patterns on linear families of polynomials over a finite field,” Combinatorica, vol. 37, no. 5, pp. 805–836, 2017.
[14] J. Gmainer and H. Havlicek, “Nuclei of normal rational curves,” Journal of Geometry, vol. 69, pp. 117–130, 2000.
[15] T. Wang, “Splitting of polynomial families via galois theory,” https://arxiv.org/abs/2606.12810, 2026, version 1, 11 June 2026.